INSTITUTO FEDERAL Santa Catarina Câmpus São José	INSTITUTO FEDERAL DE SANTA CATARINA CAMPUS SÃO JOSÉ ENGENHARIA DE TELECOMUNICAÇÕES CAL129003 - SEQUÊNCIAS E SÉRIES Profa. Silviana Cirino – silvianac@ifsc.edu.br

Lista de Exercícios: Séries de Potência

Resolver os exercícios ímpares a seguir:

1) Encontre o raio de convergência da série de potências dada:

1. $\sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n+1}$

2. $\sum_{n=0}^{\infty} (4x)^n$

3. $\sum_{n=1}^{\infty} \frac{(2x)^n}{n^2}$

4. $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{2^n}$

5. $\sum_{n=0}^{\infty} \frac{(2x)^n}{n!}$

6. $\sum_{n=0}^{\infty} \frac{(2n)! x^n}{n!}$

2) Encontre o intervalo de convergência da série de potências dada:

7. $\sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n$

8. $\sum_{n=0}^{\infty} \left(\frac{x}{k}\right)^n$

9. $\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n}$

10. $\sum_{n=0}^{\infty} (-1)^{n+1} n x^n$

11. $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

12. $\sum_{n=0}^{\infty} \frac{(3x)^n}{(2n)!}$

13. $\sum_{n=0}^{\infty} (2n)! \left(\frac{x}{2}\right)^n$

14. $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{(n+1)(n+2)}$

15. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{4^n}$

16. $\sum_{n=0}^{\infty} \frac{(-1)^n n! (x-4)^n}{3^n}$

17. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} (x-5)^n}{n 5^n}$

18. $\sum_{n=0}^{\infty} \frac{(x-2)^{n+1}}{(n+1) 3^{n+1}}$

19. $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} (x-1)^{n+1}}{n+1}$

20. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} (x-c)^n}{n c^n}$

21. $\sum_{n=1}^{\infty} \frac{(x-c)^{n-1}}{c^{n-1}}, \ 0 < c$

22. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n-1}}{2n-1}$

23. $\sum_{n=1}^{\infty} \frac{n}{n+1}(-2x)^{n-1}$

24. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!}$

25. $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$

26. $\sum_{n=1}^{\infty} \frac{n!x^n}{(2n)!}$

27. $\sum_{n=1}^{\infty} \frac{k(k+1)(k+2)\cdots(k+n-1)x^n}{n!}, \quad k \geq$

28. $\sum_{n=1}^{\infty} \left(\frac{2\cdot 4\cdot 6\cdots 2n}{3\cdot 5\cdot 7\cdots(2n+1)}\right)x^{2n+1}$

29. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}3\cdot 7\cdot 11\cdots(4n-1)(x-3)^n}{4^n}$

30. $\sum_{n=1}^{\infty} \frac{n!(x-c)^n}{1\cdot 3\cdot 5\cdots(2n-1)}$

3) Encontre uma expansão em séries de potências para a função dada em torno de *c* e determine o seu intervalo de convergência.

1. $f(x) = \frac{1}{2-x}, c = 0$	8. $f(x) = \frac{1}{2x-5}, c = 0$
2. $f(x) = \frac{3}{4-x}, c = 0$	9. $f(x) = \frac{3}{x+2}, c = 0$
3. $f(x) = \frac{1}{2-x}, c = 5$	10. $f(x) = \frac{4}{3x+2}, c = 2$
4. $f(x) = \frac{3}{4-x}, c = -2$	11. $f(x) = \frac{3x}{x^2+x-2}, c = 0$
5. $f(x) = \frac{3}{2x-1}, c = 0$	12. $f(x) = \frac{4x-7}{2x^2+3x-2}, c = 0$
6. $f(x) = \frac{3}{2x-1}, c = 2$	13. $f(x) = \frac{2}{1-x^2}, c = 0$
7. $f(x) = \frac{1}{2x-5}, c = -3$	14. $f(x) = \frac{4}{4+x^2}, c = 0$

4) Encontre a série de Taylor (centrada em c) da função dada:

1. $f(x) = e^{2x}$, $c = 0$ **2.** $f(x) = e^{-2x}$, $c = 0$

3. $f(x) = \cos x$, $c = \dfrac{\pi}{4}$ **4.** $f(x) = \text{sen}\, x$, $c = \dfrac{\pi}{4}$

5. $f(x) = \ln x$, $c = 1$ **6.** $f(x) = e^{x}$, $c = 1$

7. $f(x) = \text{sen}\, 2x$, $c = 0$

8. $f(x) = \text{tg}\, x$, $c = 0$ (os três primeiros termos não-nulos)

9. $f(x) = \sec x$, $c = 0$ (os três primeiros termos não-nulos)

10. $f(x) = \ln(x^2 + 1)$, $c = 0$

5) Encontre a série de Maclaurin da função dada a partir de uma das séries elementares:

17. $f(x) = e^{x^2/2}$ **18.** $g(x) = e^{-3x}$

19. $g(x) = \text{sen}\, 2x$ **20.** $h(x) = x \cos x$

21. $f(x) = \cos \sqrt{x}$ **22.** $f(x) = \text{sen}\, x^2$

23. $g(x) = \dfrac{\text{sen}\, x}{x}$ **24.** $f(x) = \dfrac{\text{arcsen}\, x}{x}$

25. $f(x) = \dfrac{1}{2}(e^{x} - e^{-x}) = \text{senh}\, x$

26. $f(x) = \dfrac{1}{2}(e^{x} + e^{-x}) = \cosh x$

6) Encontre a série de Maclaurin para a função $f(x)$ definida pela integral dada:

a)	b)	c)
$\int_0^x (e^{-t^2} - 1)\, dt$	$\int_0^x \sqrt{1 + t^3}\, dt$	$\int_0^x \frac{\text{sen}\, t}{t}\, dt$
d)	e)	f)
$\int_0^x \cos \frac{\sqrt{t}}{2}\, dt$	$\int_0^x \frac{\ln(t + 1)}{t}\, dt$	$\int_0^x \frac{e^t - 1}{t}\, dt$

Respostas:

Exercícios 1 e 2:

1. $R = 1$ **3.** $R = \frac{1}{2}$ **5.** $R = \infty$
7. $(-2, 2)$ **9.** $(-1, 1]$ **11.** $(-\infty, \infty)$
13. $x = 0$ **15.** $(-4, 4)$ **17.** $(0, 10]$
19. $(0, 2]$ **21.** $(0, 2c)$ **23.** $\left(-\frac{1}{2}, \frac{1}{2}\right)$
25. $(-\infty, \infty)$ **27.** $(-1, 1)$ **29.** $x = 3$

Exercício 3:

1. $\sum_{n=0}^{\infty} \frac{x^n}{2^{n+1}}$, $(-2, 2)$ **3.** $\sum_{n=0}^{\infty} \frac{(x-5)^n}{(-3)^{n+1}}$, $(2, 8)$

5. $-3 \sum_{n=0}^{\infty} (2x)^n$, $\left(-\frac{1}{2}, \frac{1}{2}\right)$

7. $-\frac{1}{11} \sum_{n=0}^{\infty} \left[\frac{2}{11}(x+3)\right]^n$, $\left(-\frac{17}{2}, \frac{5}{2}\right)$

9. $\frac{3}{2} \sum_{n=0}^{\infty} \left(\frac{x}{-2}\right)^n$, $(-2, 2)$

11. $\sum_{n=0}^{\infty} \left[\frac{1}{(-2)^n} - 1\right] x^n$, $(-1, 1)$

13. $2 \sum_{n=0}^{\infty} x^{2n}$, $(-1, 1)$

Exercício 4 e 5:

1. $\sum_{n=0}^{\infty} \frac{(2x)^n}{n!}$ **3.** $\frac{\sqrt{2}}{2} \sum_{n=0}^{\infty} \frac{(-1)^{n(n+1)/2}}{n!} \left(x - \frac{\pi}{4}\right)^n$

5. $\sum_{n=0}^{\infty} \frac{(-1)^n (x-1)^{n+1}}{n+1}$ **7.** $\sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n+1}}{(2n+1)!}$

9. $1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \cdots$

11. $\sum_{n=0}^{\infty} (-1)^n (n+1) x^n$

13. $\frac{1}{2}\left[1 + \sum_{n=1}^{\infty} \frac{(-1)^n 1 \cdot 3 \cdot 5 \cdots (2n-1) x^{2n}}{2^{3n} n!}\right]$

15. $1 + \frac{x^2}{2} + \sum_{n=2}^{\infty} \frac{(-1)^{n+1} 1 \cdot 3 \cdot 5 \cdots (2n-3) x^{2n}}{2^n n!}$

17. $1 + \frac{x^2}{2} + \frac{x^4}{2^2 2!} + \frac{x^6}{2^3 3!} + \cdots$

19. $\sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n+1}}{(2n+1)!}$ 21. $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{(2n)!}$

23. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!}$ 25. $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$

Exercício 6:

a)	c)	e)
$\sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{2n+3}}{(2n+3)(n+1)!}$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n+1)!}$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{(n+1)^2}$

Fonte: Larson, R.; Hostetler, R.; Edwards, B.. Cálculo com aplicações. V.2, 5ª Ed.. Rio de Janeiro: LTC: 1998.